Electrical Transient Analyzer Program

Load Flow Analysis

Loading Category (1): Design

Generation Category (1): Design

Load Diversity Factor: None

	Swing	V-Control	Load	Total
Number of Buses:	1	0	3	4

	XFMR2	XFMR3	Reactor	Line/Cable/ Busway	Impedance	Tie PD	Total
Number of Branches:	1	0	0	1	0	1	3

Method of Solution:	Adaptive Newton-Raphson Method
Maximum No. of Iteration:	99
Precision of Solution:	0.0001000
System Frequency:	60.00 Hz
Unit System:	English
Project Filename:	Project1
Output Filename:	C:\ETAP Course\ETAP projects\Project1\Project1\study1.lfr

Adjustments

Tolerance	Apply Adjustments	Individual /Global	Percent
Transformer Impedance:	Yes	Individual	
Reactor Impedance:	Yes	Individual	
Overload Heater Resistance:	No		
Transmission Line Length:	No		
Cable / Busway Length:	No		

Temperature Correction	Apply Adjustments	Individual /Global	Degree C
Transmission Line Resistance:	Yes	Individual	
Cable / Busway Resistance:	Yes	Individual	

Bus Input Data

Bus			Initial Voltage		Load							
					Constant kVA		Constant Z		Constant I		Generic	
ID	kV	Sub-sys	% Mag.	Ang.	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar
Bus1	11.000	1	100.0	0.0								
Bus2	11.000	1	100.0	0.0								
Bus3	0.400	1	100.0	0.0								
Bus4	0.400	1	100.0	0.0	0.845	0.357	0.644	0.274				
Total Number of Buses: 4					0.845	0.357	0.644	0.274	0.000	0.000	0.000	0.000

Generation Bus				Voltage		Generation			Mvar Limits	
ID	kV	Type	Sub-sys	% Mag.	Angle	MW	Mvar	% PF	Max	Min
Bus1	11.000	Swing	1	100.0	0.0					
						0.000	0.000			

Line/Cable/Busway Input Data

Line/Cable/Busway	ohms or siemens/1000 ft per Conductor (Cable) or per Phase (Line/Busway)								
			Length						
ID	Library	Size	Adj. (ft)	% Tol.	#/Phase	T (°C)	R	X	Y
Cable1	15NCUS3	95	328.1	0.0	1	75	0.071806	0.037490	

Line / Cable / Busway resistances are listed at the specified temperatures.

2-Winding Transformer Input Data

Transformer		Rating					Z Variation			% Tap Setting		Adjusted	Phase Shift	
ID	Phase	MVA	Prim. kV	Sec. kV	% Z1	X1/R1	+ 5%	- 5%	% Tol.	Prim.	Sec.	% Z	Type	Angle
TRANS	3-Phase	2.000	11.000	0.400	6.00	7.10	0	0	0	0	0	6.0000	Dyn	0.000

Branch Connections

CKT/Branch		Connected Bus ID		% Impedance, Pos. Seq., 100 MVA Base			
ID	Type	From Bus	To Bus	R	X	Z	Y
TRANS	2W XFMR	Bus2	Bus3	41.85	297.07	300.00	
BusDuct1	Bus Duct	Bus3	Bus4				
Cable1	Cable	Bus1	Bus2	1.95	1.02	2.20	

LOAD FLOW REPORT

Bus		Voltage		Generation		Load		Load Flow					XFMR
ID	kV	% Mag.	Ang.	MW	Mvar	MW	Mvar	ID	MW	Mvar	Amp	%PF	%Tap
* Bus1	11.000	100.000	0.0	1.468	0.696	0.000	0.000	Bus2	1.468	0.696	85.2	90.4	
Bus2	11.000	99.964	0.0	0.000	0.000	0.000	0.000	Bus1	-1.467	-0.696	85.2	90.4	
								Bus3	1.467	0.696	85.2	90.4	
Bus3	0.400	97.368	-2.4	0.000	0.000	0.000	0.000	Bus2	-1.456	-0.617	2344.3	92.1	
								Bus4	1.456	0.617	2344.3	92.1	
Bus4	0.400	97.368	-2.4	0.000	0.000	1.456	0.617	Bus3	-1.456	-0.617	2344.3	92.1	

* Indicates a voltage regulated bus (voltage controlled or swing type machine connected to it)

# Indicates a bus with a load mismatch of more than 0.1 MVA

Bus Loading Summary Report

Bus			Directly Connected Load								Total Bus Load			
			Constant kVA		Constant Z		Constant I		Generic					
ID	kV	Rated Amp	MW	Mvar	MW	Mvar	MW	Mvar	MW	Mvar	MVA	% PF	Amp	Percent Loading
Bus1	11.000										1.624	90.4	85.2	
Bus2	11.000										1.624	90.4	85.2	
Bus3	0.400										1.581	92.1	2344.3	
Bus4	0.400		0.845	0.357	0.611	0.260					1.581	92.1	2344.3	

* Indicates operating load of a bus exceeds the bus critical limit (100.0% of the Continuous Ampere rating).

# Indicates operating load of a bus exceeds the bus marginal limit (95.0% of the Continuous Ampere rating).

Branch Loading Summary Report

CKT / Branch		Busway / Cable & Reactor			Transformer				
						Loading (input)		Loading (output)	
ID	Type	Ampacity (Amp)	Loading Amp	%	Capability (MVA)	MVA	%	MVA	%
Cable1	Cable	236.13	85.25	36.10					
TRANS	Transformer				2.000	1.624	81.2	1.581	79.1

* Indicates a branch with operating load exceeding the branch capability.

Branch Losses Summary Report

Branch ID	From-To Bus Flow		To-From Bus Flow		Losses		% Bus Voltage		Vd % Drop
	MW	Mvar	MW	Mvar	kW	kvar	From	To	in Vmag
Cable1	1.468	0.696	-1.467	-0.696	0.5	0.3	100.0	100.0	0.04
TRANS	1.467	0.696	-1.456	-0.617	11.0	78.4	100.0	97.4	2.60
					11.6	78.6			

* This Transmission Line includes Series Capacitor.

Alert Summary Report

	% Alert Settings	
	Critical	**Marginal**
Loading		
Bus	100.0	95.0
Cable / Busway	100.0	95.0
Reactor	100.0	95.0
Line	100.0	95.0
Transformer	100.0	95.0
Panel	100.0	95.0
Protective Device	100.0	95.0
Generator	100.0	95.0
Inverter/Charger	100.0	95.0
Bus Voltage		
OverVoltage	105.0	102.0
UnderVoltage	95.0	98.0
Generator Excitation		
OverExcited (Q Max.)	100.0	95.0
UnderExcited (Q Min.)	100.0	

Marginal Report

Device ID	Type	Condition	Rating/Limit	Unit	Operating	% Operating	Phase Type
Bus3	Bus	Under Voltage	0.400	kV	0.389	97.4	3-Phase
Bus4	Bus	Under Voltage	0.400	kV	0.389	97.4	3-Phase

SUMMARY OF TOTAL GENERATION , LOADING & DEMAND

	MW	Mvar	MVA	% PF
Source (Swing Buses):	1.468	0.696	1.624	90.36 Lagging
Source (Non-Swing Buses):	0.000	0.000	0.000	
Total Demand:	1.468	0.696	1.624	90.36 Lagging
Total Motor Load:	0.845	0.357	0.918	92.12 Lagging
Total Static Load:	0.611	0.260	0.664	92.00 Lagging
Total Constant I Load:	0.000	0.000	0.000	
Total Generic Load:	0.000	0.000	0.000	
Apparent Losses:	0.012	0.079		
System Mismatch:	0.000	0.000		

Number of Iterations: 2